TUTORIAL 9

Dimension Formulas, Row and Column Spaces, and Rank

NOTATION AND CONVENTIONS

For $A \in M_{m\times n}(F)$, the row space $\operatorname{Row} A \leq F^n$ is the span of its rows, and the column space $\operatorname{Col} A \leq F^m$ is the span of its columns. Rows are viewed as vectors of F^n. The symbol $\dim U$ denotes the dimension of a subspace U. The rank $\operatorname{rank} A$ is the number of pivots in reduced row echelon form; below we show that it equals both the row-space and column-space dimensions.

DEFINITION 9.1

A precise formulation with hypotheses. For $S \subseteq V$, its span consists of all finite sums $\sum_{j=1}^{k} \alpha_j v_j$ with $v_j \in S$ and $\alpha_j \in \mathbb{F}$, including the empty sum, which equals 0_V. It is the smallest subspace containing S. Independence means that every finite zero linear combination of distinct members of S has only zero coefficients. In particular, the empty set is independent and its span is $\{0_V\}$.
For subspaces $U, W \leq V$ over the same field, $U+W$ is the smallest subspace containing both. The condition $U \cap W = \{0_V\}$ is equivalent to every $x \in U+W$ having a unique expression $x = u + w$ with $u \in U, w \in W$. Indeed, two expressions give $u - u' = w' - w \in U \cap W$. Conversely, if $z \in U \cap W$, then $z = z + 0_V = 0_V + z$, so uniqueness forces $z = 0_V$.

MOTIVATION AND GUIDING IDEA

Motivation. Matrix multiplication describes composition of linear operations: in ABx, apply B first and then A. Checking dimensions reveals in advance whether the composition is possible. A basis gives a unique description of every vector, and dimension counts the coefficients required for that description.

NOTATION AND CONVENTIONS

Notation guide. $M_{m\times n}(\mathbb{F})$ is the space of matrices with m rows and n columns; a_{ij} is the entry in row i, column j. $I_n = (\delta_{ij})$, where $\delta_{ij} = 1$ if $i = j$ and is zero otherwise. e_j is the jth standard basis vector, and Ae_j is column j of A. A^T denotes transpose, $\operatorname{diag}(d_1, \ldots, d_n)$ a diagonal matrix, $|B|$ the cardinality of a finite set, and $\dim_{\mathbb{F}} V$ the dimension over $\mathbb{F}$.

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DEFINITION 9.2

Hypotheses to check. If $A \in M_{k\times m}(\mathbb{F})$ and $B \in M_{m\times n}(\mathbb{F})$, then $AB \in M_{k\times n}(\mathbb{F})$ with $(AB)_{ij} = \sum_{r=1}^{m} a_{ir}b_{rj}$. Also $(AB)^T = B^T A^T$. Over $\mathbb{C}$, the transpose A^T does not include complex conjugation.
For square matrices, define $\mathrm{Sym}_n(\mathbb{F}) = \{A : A^T = A\}$ and $\mathrm{ASym}_n(\mathbb{F}) = \{A : A^T = -A\}$. Both sets are subspaces over any field. If the characteristic is different from 2, a skew-symmetric matrix has zero diagonal and every matrix has a unique symmetric/skew-symmetric decomposition using division by 2. In characteristic 2 the two conditions coincide and do not imply a zero diagonal.
If $\dim_{\mathbb{F}} V = n < \infty$, a set of n vectors is a basis if and only if it is independent, and if and only if it spans. For $U, W \le V$, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$; subtracting the intersection dimension prevents shared directions from being counted twice.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Compute with matrices while checking dimensions and multiplication order, recognize symmetric structure, and choose bases. Connect algebraic calculations with sums and intersections of subspaces.

9.1 Intersections and Sums of Subspaces

MOTIVATION AND GUIDING IDEA

Distinguish union from sum. For example, the two coordinate axes in $\mathbb{R}^2$ are subspaces, but their union is not closed under addition: $(2, 0)$ and $(0, -3)$ belong to the union, whereas their sum $(2, -3)$ does not. The sum of the two axes is the whole plane.

RECALL AND TOOLS

If $U, W \subseteq V$ are subspaces of a vector space V, their **intersection** is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 9.3

For subspaces $U, W \subseteq V$, define their **sum** by

$$U + W = \{u + w : u \in U, w \in W\}.$$

Then $U + W$ is a subspace of V containing both U and W. If also $U \cap W = \{0\}$, the sum is called a **direct sum** and is denoted $U \oplus W$.

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EXERCISE 9.1

Please write

$$\text{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \text{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}$$

for the symmetric and skew-symmetric matrices. Please prove that both sets are subspaces of $M_n(\mathbb{F})$. In addition, please prove that if $\text{char}(\mathbb{F}) \neq 2$, that is, $1 + 1 \neq 0$ in the field, then

$$M_n(\mathbb{F}) = \text{Sym}(\mathbb{F}) \oplus \text{aSym}(\mathbb{F}).$$

WORKED SOLUTION

We verify that $\text{Sym}(\mathbb{F})$ is a subspace. For $\text{aSym}(\mathbb{F})$, the zero matrix belongs to the set, and if $A^T = -A$, $B^T = -B$, and $\lambda \in \mathbb{F}$, then

$$(A + B)^T = A^T + B^T = -A - B = -(A + B), \qquad (\lambda A)^T = \lambda A^T = -\lambda A.$$

Thus skew-symmetric matrices are also closed under addition and scalar multiplication and form a subspace.
Now let $A, B \in \text{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. First, $0 \in \text{Sym}(\mathbb{F})$. For addition, $(A + B)^T = A^T + B^T = A + B$, so $A + B$ is symmetric. For scalar multiplication, $(\lambda A)^T = \lambda A^T = \lambda A$, so λA is symmetric. To prove the direct-sum assertion, first prove that the sum is the whole matrix space and then prove that the intersection is zero. For the equality $M_n(\mathbb{F}) = \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$, check both inclusions:

1. $M_n(\mathbb{F}) \supseteq \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$ holds because both summands are subspaces of $M_n(\mathbb{F})$.
2. For the reverse inclusion, take $A \in M_n(\mathbb{F})$ and write

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric, since

$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2},$$

and the second is skew-symmetric, since

$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$

Hence $M_n(\mathbb{F}) \subseteq \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$.

Finally, suppose $A \in \text{Sym}(\mathbb{F}) \cap \text{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, while skew-symmetry gives $-A = A^T$. Thus $A = -A$, and $a_{ij} = -a_{ij}$ for every $1 \leq i, j \leq n$. Consequently $a_{ij} + a_{ij} = 2a_{ij} = 0$. Because $\text{char}(\mathbb{F}) \neq 2$, the scalar $2 = 1 + 1$ is nonzero and invertible in the field. Therefore every $a_{ij} = 0$, so $A = 0$, as required.

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RECALL AND TOOLS

In characteristic 2, $-A = A$, so the conditions $A^T = A$ and $A^T = -A$ coincide. This is exactly why the symmetric/skew-symmetric decomposition using division by 2 requires characteristic different from 2.

9.2 Intersections and Sums of Subspaces

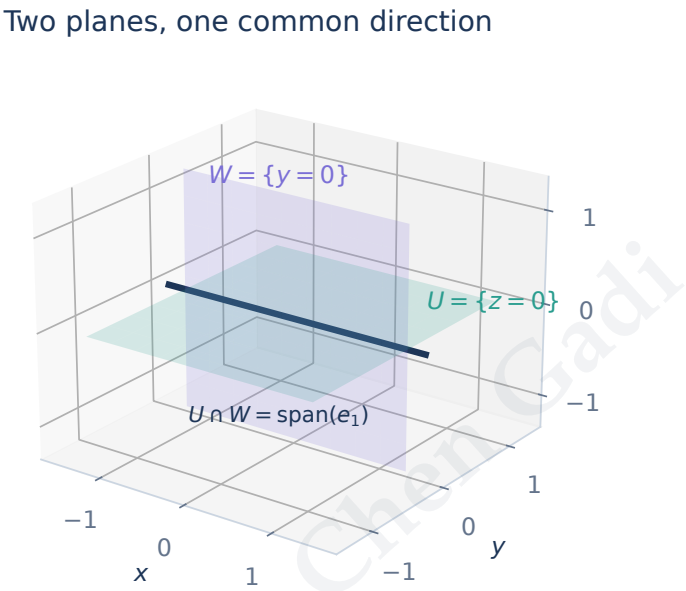


Figure 9.1. Two distinct planes through the origin in $\mathbb{R}^3$ may intersect in a line. Here $U = \{z = 0\}$ and $W = \{y = 0\}$, so $U \cap W = \text{span}\{(1, 0, 0)\}$ and $U + W = \mathbb{R}^3$.

RECALL AND TOOLS

For subspaces $U, W \subseteq V$ of a vector space V, their intersection is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 9.4

For subspaces $U, W \subseteq V$, define

$$U + W = \{u + w : u \in U, w \in W\}.$$

This is a subspace of V containing U and W. If also $U \cap W = \{0\}$, the sum is a **direct sum**, denoted $U \oplus W$.

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EXERCISE 9.2

Let

$$\mathrm{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \mathrm{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}.$$

Please prove that the symmetric and skew-symmetric matrices are subspaces of $M_n(\mathbb{F})$, and that whenever $\mathrm{char}(\mathbb{F}) \neq 2$,

$$M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) \oplus \mathrm{aSym}(\mathbb{F}).$$

WORKED SOLUTION

We show that $\mathrm{Sym}(\mathbb{F})$ is a subspace. For the skew-symmetric set, the corresponding identities are

$$0^T = -0, \quad (A+B)^T = -A - B = -(A+B), \quad (\lambda A)^T = -\lambda A,$$

so this set is also a subspace. For the symmetric set, take $A, B \in \mathrm{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Clearly $0 \in \mathrm{Sym}(\mathbb{F})$. Closure under addition follows from $(A+B)^T = A^T + B^T = A + B$. Closure under scalar multiplication follows from $(\lambda A)^T = \lambda A^T = \lambda A$. To establish the direct sum, first prove equality with the sum and then show that the intersection is zero. Check both inclusions for $M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$:

1. $M_n(\mathbb{F}) \supseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$ follows because both summands are subspaces of $M_n(\mathbb{F})$.
2. Take $A \in M_n(\mathbb{F})$. Then
$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$
The first summand is symmetric because
$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2}.$$
The second is skew-symmetric because
$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$
This proves the reverse inclusion.

Now take $A \in \mathrm{Sym}(\mathbb{F}) \cap \mathrm{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, and skew-symmetry gives $-A = A^T$. Therefore $A = -A$, so $a_{ij} = -a_{ij}$ for every $1 \leq i, j \leq n$. Thus $a_{ij} + a_{ij} = 2a_{ij} = 0$. As the field has characteristic different from 2, the scalar 2 is nonzero, so every $a_{ij} = 0$. Hence $A = 0$, as required.

The dimension formula:

PROPOSITION 9.1

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

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EXERCISE 9.3

1. Let V have dimension n, and let U, W be subspaces with $\dim U = n-1$ and $W \not\subseteq U$. Please prove $V = U + W$ and $\dim(U \cap W) = \dim W - 1$.
2. $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2+4\}$, $W = \{p \in \mathbb{R}_2[X] : p'(2) = 0\}$. Please compute the dimensions of the sum and intersection, and please determine whether the sum equals V and is direct.

WORKED SOLUTION

1. Choose $w \in W \setminus U$. Then $U \subsetneq U + W \subseteq V$, so

$$n - 1 = \dim U < \dim(U + W) \leq n.$$

Thus $\dim(U + W) = n$ and $U + W = V$. The dimension formula gives

$$n = (n-1) + \dim W - \dim(U \cap W),$$

hence $\dim(U \cap W) = \dim W - 1$.

2. The generators of U are independent, so $\dim U = 2$. Also $1 \in W \setminus U$. For W,

$$p(X) = aX^2 + bX + c \in W \iff 4a + b = 0 \iff b = -4a.$$

$$W = \operatorname{span}\{1, X^2 - 4X\}, \qquad \dim W = 2.$$

Part 1 gives $U + W = V$. The sum is not direct, since

$$(X-2)^2 = (X^2+4) - 4X \in U, \qquad \frac{d}{dX}(X-2)^2\Big|_{X=2} = 0.$$

$$U \cap W = \operatorname{span}\{(X-2)^2\}, \quad \dim(U \cap W) = 1, \quad \dim(U + W) = 2 + 2 - 1 = 3.$$

The dimension formula:

PROPOSITION 9.2

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 9.4

1. Let V have dimension n, and let U, W be subspaces with $\dim U = n-1$ and $W \not\subseteq U$. Please prove $V = U + W$ and $\dim(U \cap W) = \dim W - 1$.
2. $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2+9\}$, $W = \{p \in \mathbb{R}_2[X] : p'(-3) = 0\}$. Please compute the dimensions of the sum and intersection, and please determine whether the sum equals V and is direct.

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EXERCISE 9.4 CONTINUED

WORKED SOLUTION

1. Choose $w \in W \setminus U$. Then $U \subsetneq U + W \subseteq V$, so

$$n - 1 = \dim U < \dim(U + W) \leq n.$$

Thus $\dim(U + W) = n$ and $U + W = V$. The dimension formula gives

$$n = (n - 1) + \dim W - \dim(U \cap W),$$

hence $\dim(U \cap W) = \dim W - 1$.

2. The generators of U are independent, so $\dim U = 2$. Also $1 \in W \setminus U$. For W,

$$p(X) = aX^2 + bX + c \in W \iff -6a + b = 0 \iff b = 6a.$$

$$W = \text{span}\{1, X^2 + 6X\}, \qquad \dim W = 2.$$

Part 1 gives $U + W = V$. The sum is not direct, since

$$(X + 3)^2 = (X^2 + 9) + 6X \in U, \qquad \frac{d}{dX}(X + 3)^2\Big|_{X=-3} = 0.$$

$$U \cap W = \text{span}\{(X + 3)^2\}, \quad \dim(U \cap W) = 1, \quad \dim(U + W) = 2 + 2 - 1 = 3.$$

9.3 Row and column spaces and the rank theorem

RECALL AND TOOLS

The **row space** $\mathcal{R}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^n$ spanned by its rows. .

- Elementary row operations preserve the row space of a matrix.

PROPOSITION 9.3

- **Claim 1.** If $A, B \in M_{m\times n}(\mathbb{F})$ are row equivalent, then $\mathcal{R}(A) = \mathcal{R}(B)$.
- **Claim 2:** If A is in row echelon form, its nonzero rows form a basis of $\mathcal{R}(A)$, (and are therefore linearly independent in $\mathbb{F}^n$).

RECALL AND TOOLS

The **column space** $\mathcal{C}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^m$ spanned by its columns. .

- Row reduction **does not in general preserve the column space**; only its **dimension** is preserved.
- $\mathcal{C}(A) = \mathcal{R}(A^T)$ when a row vector is identified with its transposed column vector.

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PROPOSITION 9.4

The rank theorem: Every matrix A satisfies $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$. This common dimension is called the **rank of the matrix**.

RECALL AND TOOLS

- The row and column spaces need not be the same subspace. For a rectangular matrix they even lie in different ambient spaces, $\mathbb{F}^n$ and $\mathbb{F}^m$, respectively; their dimensions are nevertheless equal.
- Equivalently, rank$A$ is the number of pivots in the reduced row echelon form.
- For $A \in \mathbb{R}^{m\times n}$, $\text{rank}(A) \leq \min(m, n)$.
- A square matrix $A \in \mathbb{R}^{n\times n}$ is invertible if and only if $rank(A) = n$
- $rank(A \cdot B) \leq min\{rank(A), rank(B)\}$
- $rank(A + B) \leq rank(A) + rank(B)$

EXERCISE 9.5

$$A = \begin{pmatrix} 2 & 6 & -4 \\ 1 & 3 & -2 \end{pmatrix}.$$

Please find a basis and dimension for the row space and column space of A.

WORKED SOLUTION

Swap the rows and subtract twice the first row from the second. The column-space basis is taken from the original matrix.

$$A \sim \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$\text{Row}(A) = \text{span}\{(1, 3, -2)\}, \qquad \text{Col}(A) = \text{span}\{(2, 1)^T\}.$$

$$\text{Col}_2(A) = 3\,\text{Col}_1(A), \qquad \text{Col}_3(A) = -2\,\text{Col}_1(A), \qquad \dim \text{Row}(A) = \dim \text{Col}(A) = 1.$$

EXERCISE 9.6

Let $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times m}$ satisfy $AB = I_m$. Please find the ranks of A and B.
What can be concluded about m and n?

WORKED SOLUTION

The rank inequalities give $\text{rank}(AB) = \text{rank}(I_m) = m \leq \text{rank}(A) \leq m$.
Thus $\text{rank}(A) = m$. For B, $m = \text{rank}(AB) \leq \text{rank}(B) \leq m$, so $\text{rank}(B) = m$ as well.
Since $\text{rank}(A) = m \leq \min(m, n)$, we obtain $m \leq n$.
The matrices are not assumed square, so one cannot conclude that B is an inverse of A.

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9.3.1 Reducing spanning sets and extending independent sets to bases in $\mathbb{F}^n$

Every spanning set contains a basis, and every linearly independent set extends to a basis.

Reducing a spanning set to a basis

MOTIVATION AND GUIDING IDEA

1. **Method A:** We want a basis consisting of vectors from the original set (as in the preceding exercise).
   - Form a matrix whose **columns** are the vectors of S and reduce it to row echelon form.
   - For each pivot column, choose the corresponding **original column** as a basis vector. These columns are linearly independent.
   - The column space is not preserved by row operations; therefore the columns must be taken from the original matrix.
2. **Method B:** Find a **new** linearly independent set spanning the same space.
   - Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
   - Its nonzero echelon rows form a basis.
   - Row operations preserve the row space, so the basis may be taken after reduction.

Extending a linearly independent set to a basis

MOTIVATION AND GUIDING IDEA

- Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
- For each nonpivot column j, adjoin the standard basis vector e_j to S. To justify this construction, first replace the original rows by the echelon rows, preserving their span. At the pivot indices, the added standard basis vectors are zero, whereas the echelon rows give a triangular matrix with nonzero diagonal. Hence a vanishing linear combination first forces the coefficients of the echelon rows to be zero, and then those of the added standard basis vectors. Since the original rows are independent, replacing the echelon rows back by the originals preserves the basis property.

EXERCISE **9.7**

$$S = \{(1,2,0,1,-1)^T, (2,4,1,3,0)^T, (1,2,1,2,2)^T\} \subseteq \mathbb{R}^5.$$

Please prove that S is linearly independent and please extend it to a basis of $\mathbb{R}^5$.

WORKED SOLUTION

The three nonzero echelon rows prove independence. Pivot columns are $1, 3, 5$. Adding e_2, e_4 gives five independent rows: the pivot columns first force the three echelon-row coefficients

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EXERCISE 9.7 CONTINUED

to vanish, and columns $2, 4$ then force the remaining two coefficients to vanish.

$$\begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 4 & 1 & 3 & 0 \\ 1 & 2 & 1 & 2 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$B = S \cup \{e_2, e_4\}.$$

EXERCISE 9.8

$$V = \operatorname{span}\{(1,3,2,-1)^T, (2,7,5,0)^T, (0,1,1,2)^T\} \subseteq \mathbb{R}^4.$$

Please find a basis and dimension for V and please extend that basis to a basis of $\mathbb{R}^4$.

WORKED SOLUTION

The first two coordinates show that the two vectors in B are independent. They span all three given vectors. The final determinant verifies the extension.

$$(2,7,5,0)^T = 2(1,3,2,-1)^T + (0,1,1,2)^T.$$

$$B = \{(1,3,2,-1)^T, (0,1,1,2)^T\}, \qquad \dim V = 2.$$

$$\det \begin{pmatrix} 1 & 3 & 2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 1, \qquad D = B \cup \{e_3, e_4\}.$$

EXERCISE 9.9 CHALLENGE

Let U, W be subspaces of a vector space V over an arbitrary field $\mathbb{F}$. Please prove that the following are equivalent:

1. $U \cup W$ is a subspace of V.
2. $U \subseteq W$ or $W \subseteq U$.
3. $U + W = U \cup W$.

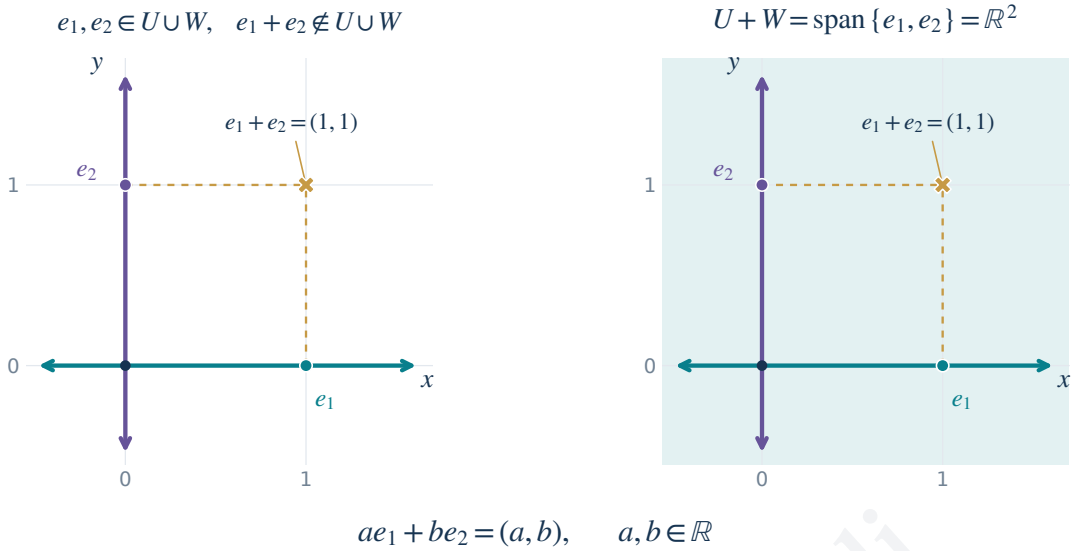
WORKED SOLUTION

(1) $\Rightarrow$ (2): Suppose neither inclusion holds. Choose $u \in U \setminus W$ and $w \in W \setminus U$. Both belong to the union, so closure under addition gives $u + w \in U \cup W$. If $u + w \in U$, then $w = (u + w) - u \in U$, a contradiction. If $u + w \in W$, then $u = (u + w) - w \in W$, again a contradiction. Hence one inclusion must hold.
(2) $\Rightarrow$ (3): If $U \subseteq W$, then $U \cup W = W$ and $U + W = W$: every such sum belongs to W, and each $w \in W$ equals $0_V + w$. The other case follows by interchanging the subspaces.
(3) $\Rightarrow$ (1): The sum $U + W$ is a subspace, so the equal set $U \cup W$ is also a subspace. The proof divides by no scalar and therefore applies to finite fields and characteristic 2 as well.

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$$U = \mathbb{R}e_1, \qquad W = \mathbb{R}e_2, \qquad e_1 = (1,0), \quad e_2 = (0,1)$$



$$ae_1 + be_2 = (a,b), \qquad a,b \in \mathbb{R}$$

Figure 9.2. An illustration over $\mathbb{R}$: the coordinate axes $U = \mathbb{R}e_1$ and $W = \mathbb{R}e_2$ are subspaces. Their union contains e_1 and e_2 but excludes their sum $(1,1)$, so it is not closed under addition. In contrast, every $(a,b) \in \mathbb{R}^2$ equals $ae_1 + be_2$, giving $U + W = \mathbb{R}^2$. Shading in the right panel represents the full plane within the viewing window.

9.4 Exercises

EXERCISE 9.10

Let V be a vector space over a field $\mathbb{F}$, and let $S, R \subseteq V$ be arbitrary subsets, not necessarily finite.
Please write

$$\mathrm{span}(S) = \left\{ \sum_{i=1}^{k} \alpha_i s_i \;\middle|\; k \in \mathbb{Z}_{\geq 0},\ \alpha_i \in \mathbb{F},\ s_i \in S \right\},$$

The empty sum is the zero vector, so $\mathrm{span}(\varnothing) = \{0\}$. Define $\mathrm{span}(R)$ and $\mathrm{span}(S \cup R)$ similarly.
Please show that, for all $S, R \subseteq V$,

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R).$$

WORKED SOLUTION

Write

$$U := \mathrm{span}(S), \qquad W := \mathrm{span}(R).$$

Recall that the sum $U + W$ is defined by

$$U + W := \{u + w \mid u \in U,\ w \in W\}.$$

We prove

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R)$$

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EXERCISE 9.10 CONTINUED

by establishing both inclusions:

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R) \quad \text{and} \quad \text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

1. First prove
$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$
Let $v \in \text{span}(S) + \text{span}(R)$. Then $v = u + w$ for some $u \in \text{span}(S)$ and $w \in \text{span}(R)$:
$$v = u + w.$$
By the definition of $\text{span}(S)$, there are vectors $s_1, \ldots, s_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$ such that
$$u = \sum_{i=1}^{k} \alpha_i s_i.$$
Similarly, there are vectors $r_1, \ldots, r_\ell \in R$ and scalars $\beta_1, \ldots, \beta_\ell \in \mathbb{F}$ such that
$$w = \sum_{j=1}^{\ell} \beta_j r_j.$$
Therefore,
$$v = u + w = \sum_{i=1}^{k} \alpha_i s_i + \sum_{j=1}^{\ell} \beta_j r_j.$$
Every s_i and every r_j lies in $S \cup R$. Thus v is a finite linear combination of vectors in $S \cup R$.
By the definition of span,
$$v \in \text{span}(S \cup R).$$
Hence
$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$
2. For the reverse inclusion, prove
$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$
Let $v \in \text{span}(S \cup R)$. There are $x_1, \ldots, x_t \in S \cup R$ and $\gamma_1, \ldots, \gamma_t \in \mathbb{F}$ such that
$$v = \sum_{i=1}^{t} \gamma_i x_i.$$
Partition the indices into disjoint sets: assign to S every vector belonging to S, and assign the rest to R. A vector in the intersection is counted only in the first set.
Define
$$I_S := \{i \in \{1, \ldots, t\} \mid x_i \in S\}, \qquad I_R := \{1, \ldots, t\} \setminus I_S.$$
Then v can be written as
$$v = \sum_{i \in I_S} \gamma_i x_i + \sum_{i \in I_R} \gamma_i x_i.$$

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EXERCISE 9.10 CONTINUED

Set

$$u := \sum_{i \in I_S} \gamma_i x_i, \qquad w := \sum_{i \in I_R} \gamma_i x_i.$$

Observe that

- for $i \in I_S$, $x_i \in S$, so u is a linear combination of vectors in S; thus

$$u \in \text{span}(S).$$

- for $i \in I_R$, $x_i \in R$, so w is a linear combination of vectors in R; thus

$$w \in \text{span}(R).$$

We have obtained

$$v = u + w$$

with $u \in \text{span}(S)$ and $w \in \text{span}(R)$, hence

$$v \in \text{span}(S) + \text{span}(R).$$

This proves

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

The two inclusions give

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R),$$

as required.

EXERCISE 9.11

$$U = \text{span}\{1, x, 2 + x\}, \qquad W = \{p \in \mathbb{R}_3[x] : p(-1) = p(2) = 0\}.$$

Please find bases and dimensions for $U + W$ and $U \cap W$.

WORKED SOLUTION

A polynomial of degree at most one with two distinct roots is zero. Thus the intersection is zero and the sum is direct. The four listed polynomials are independent by their degrees and span the ambient space. The intersection has the empty basis.

$$U = \text{span}\{1, x\}, \qquad p \in W \iff p = (x + 1)(x - 2)(a + bx).$$

$$W = \text{span}\{x^2 - x - 2,\ x^3 - x^2 - 2x\}, \qquad \dim W = 2.$$

$$U \cap W = \{0\}, \quad \dim(U \cap W) = 0, \qquad U + W = \mathbb{R}_3[x], \quad \dim(U + W) = 4.$$

$$B_{U+W} = (1, x, x^2 - x - 2, x^3 - x^2 - 2x).$$

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EXERCISE 9.12

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & -1 & 5 \end{pmatrix}, \qquad y = (4, -1, 3, 10)^T.$$

Please find row-space and column-space bases and the number of dependent rows. Please decide whether $Mx = y$ is solvable and whether an existing solution is unique. Please replace the fourth entry of y by 9 and please find every solution.

WORKED SOLUTION

The first two rows are independent and the other two depend on them. The third column is twice the first minus the second, so the listed columns form a basis. The row relations give necessary right-hand-side conditions; they are sufficient because the first two equations can be solved for any two right-hand sides. The original right-hand side is inconsistent. After correction, setting $x_3 = t$ gives the displayed solutions. The one-dimensional kernel prevents uniqueness.

$$R_3 = R_1 + R_2, \qquad R_4 = 2R_1 - R_2, \qquad \operatorname{rank} M = 2.$$

$$B_{\text{row}} = \{(1, 0, 2), (0, 1, -1)\}, \quad B_{\text{col}} = \{(1, 0, 1, 2)^T, (0, 1, 1, -1)^T\}.$$

$$y \in \operatorname{Col} M \iff y_3 = y_1 + y_2, \quad y_4 = 2y_1 - y_2.$$

$$2 \cdot 4 - (-1) = 9 \neq 10, \qquad \ker M = \operatorname{span}\{(-2, 1, 1)^T\}.$$

$$x = (4, -1, 0)^T + t(-2, 1, 1)^T, \qquad t \in \mathbb{R}.$$

Additional practice

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EXERCISE 9.13 CHALLENGE

Let $A \in M_n(F)$ over any field. Please prove

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) \geq n,$$

with equality if and only if $A^2 = A$.

WORKED SOLUTION

Put $U = \operatorname{Col} A$ and $W = \operatorname{Col}(I_n - A)$. Since $x = Ax + (I_n - A)x$, we have $U + W = F^n$. The dimension formula gives

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) = n + \dim(U \cap W) \geq n.$$

If equality holds, then $U \cap W = \{0\}$. For every x, the vector $A(I_n - A)x = (I_n - A)Ax$ belongs to both spaces, so it is zero. Hence $A^2 = A$. Conversely, assume $A^2 = A$ and let $y = Au = (I_n - A)v$. The first representation gives $Ay = y$, while the second gives $Ay = (A - A^2)v = 0$. Therefore $y = 0$, the intersection is trivial, and equality follows.

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EXERCISE 9.14 CHALLENGE

Let $m, n \geq 2$ and let $W \neq \{0\}$ be a subspace of $M_{m\times n}(F)$ in which every matrix has rank at most 1. Please prove that at least one of the following holds:

1. There is a fixed $u \in F^m \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $v \in F^n$.
2. There is a fixed $v \in F^n \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $u \in F^m$.

WORKED SOLUTION

Every nonzero rank-one matrix is uv^T with $u, v \neq 0$: choose a nonzero column u and collect the scalar multiples expressing the other columns into v^T. First consider uv^T and rs^T. If both u, r and v, s are independent pairs, the matrix with rows v^T, s^T has rank 2 and maps onto F^2. Thus there are x, y with

$$v^T x = 1,\ s^T x = 0, \qquad v^T y = 0,\ s^T y = 1.$$

The sum sends x to u and y to r, so its rank is at least 2. Consequently a sum of rank at most 1 forces dependence of at least one pair of factors. Choose $A_0 = u_0 v_0^T \in W$ nonzero. If every column space lies in $\text{span}\{u_0\}$, the first alternative holds. Otherwise choose B whose column direction is independent of u_0. Since $A_0 + B \in W$, the preceding argument forces the row factor of B to be a multiple of v_0. Absorbing that nonzero scalar into its column factor gives $B = rv_0^T$ with r, u_0 independent. For any nonzero $C = cd^T \in W$, suppose d, v_0 were independent. Applying the same argument to $A_0 + C$ forces $c \in \text{span}\{u_0\}$; applying it to $B + C$ forces $c \in \text{span}\{r\}$. This gives $c = 0$, a contradiction. Every row factor is therefore a multiple of v_0, proving the second alternative.

EXERCISE 9.15

Let A, B be square matrices such that AB is invertible. Please prove that both A and B are invertible. **This is the converse of the product assertion from the lecture.**

WORKED SOLUTION

Use the fact that for square matrices of the same order, a one-sided inverse is also a two-sided inverse. Indeed, $AX = I$ implies $n = \text{rank}(AX) \leq \text{rank}(A) \leq n$, so A is invertible. From $(AB)(AB)^{-1} = I$, obtain $A[B(AB)^{-1}] = I$. Thus A is invertible, with inverse $B(AB)^{-1}$. Similarly, $(AB)^{-1}(AB) = I$ gives $[(AB)^{-1}A]B = I$. Thus B is invertible, with inverse $(AB)^{-1}A$.

EXERCISE 9.16 EXAM QUESTION

Work over $\mathbb{Z}_3 = \{0, 1, 2\}$, with all arithmetic modulo 3. For each $\omega \in \mathbb{Z}_3$, define

$$A(\omega) = \begin{pmatrix} 1 & \omega & 1 \\ \omega & 1 & 1 \\ 1 & 1 & \omega \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad c = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Please determine the values of ω for which $b \in \text{Col}(A(\omega))$, and please find all solutions of $A(\omega)x = b$ in those cases.
2. Please determine the values of ω for which both $b \in \text{Col}(A(\omega))$ and $c \in \text{Col}(A(\omega))$ hold.

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EXERCISE 9.16 EXAM QUESTION CONTINUED

WORKED SOLUTION

All operations below are modulo 3. For $\omega = 0, 2$ the matrix is invertible, so the displayed solutions for b are unique. For $\omega = 1$ its column space consists of equal-entry vectors, so the system with b is inconsistent. Both parts therefore have the answer $\omega \in \{0, 2\}$. The last products also give c in each of these two cases.

$$A(0) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A(1) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad A(2) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

$$\det A(\omega) = -\omega^3 + 3\omega - 2, \qquad \det A(0) = 1, \quad \det A(2) = 2 \quad \text{in } \mathbb{Z}_3.$$

$$A(0)\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = b, \qquad A(2)\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = b.$$

$$\mathrm{Col}(A(1)) = \{(a, a, a)^T : a \in \mathbb{Z}_3\}, \qquad b \notin \mathrm{Col}(A(1)).$$

$$A(0)\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = c, \qquad A(2)\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = c.$$

EXERCISE 9.17

$$m = Cq, \qquad C = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}.$$

Please find a column-space basis. Please prove that C is invertible and please calculate C^{-1} and q in terms of m.

WORKED SOLUTION

The matrix is triangular with diagonal entries 1, so it is invertible and its columns form a basis. Solving the equations in order gives the inverse formula.

$$\det C = 1, \qquad B_{\text{col}} = \{(1, 2, -1)^T, (0, 1, 3)^T, (0, 0, 1)^T\}.$$

$$m_1 = q_1, \quad m_2 = 2q_1 + q_2, \quad m_3 = -q_1 + 3q_2 + q_3.$$

$$q_1 = m_1, \quad q_2 = m_2 - 2m_1, \quad q_3 = m_3 - 3m_2 + 7m_1.$$

$$C^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -3 & 1 \end{pmatrix}, \qquad CC^{-1} = C^{-1}C = I_3.$$

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EXERCISE 9.18 CHALLENGE

Let $n \geq 2$. Let W consist of all real $n \times n$ matrices whose row sums and column sums all equal the same number. Please prove that W is a subspace and find a basis and its dimension. You may use E_{ij}, the matrix with a single 1 in position (i, j).

WORKED SOLUTION

Addition adds the common sums, scalar multiplication scales them, and zero belongs to W. Hence W is a subspace. Let W_0 be the matrices with all row and column sums zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn}.$$

Each lies in W_0. Every $A \in W_0$ satisfies

$$A = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} a_{ij} F_{ij}.$$

The upper left $(n-1) \times (n-1)$ blocks agree. Zero row sums then determine the last column, zero column sums determine the last row, and the final entry is determined as well. These matrices are independent: entry (i, j) with $i, j < n$ reads off the coefficient of F_{ij}. If the common sum of $A \in W$ is c, then $A - cI_n \in W_0$. Since $I_n \notin W_0$, a basis is

$$\{I_n\} \cup \{F_{ij} : 1 \leq i, j < n\}, \qquad \dim W = 1 + (n-1)^2.$$

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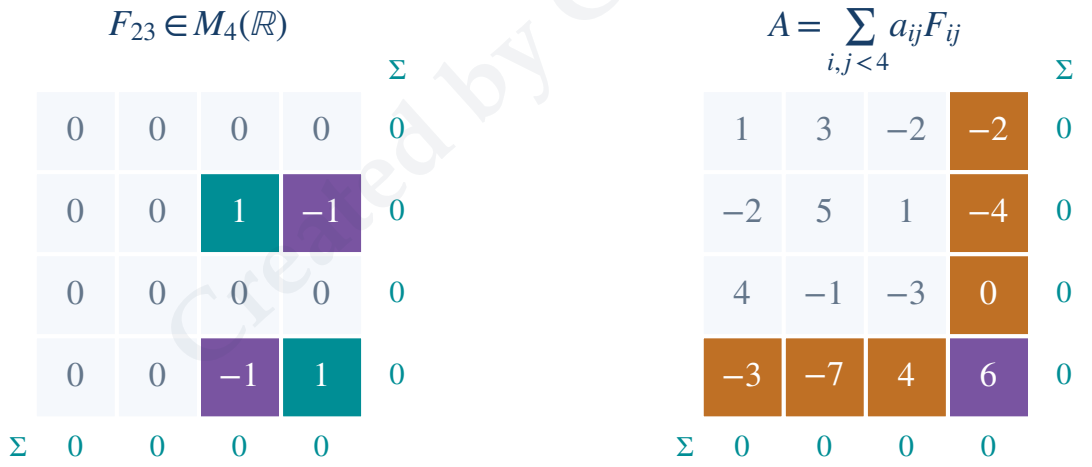


Figure 9.3. Left: the matrix F_{23} for $n = 4$. Right: an interior block completed so that every row and column sum is zero.

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EXERCISE 9.19 CHALLENGE

A permutation matrix has exactly one entry equal to 1 in each row and each column, with all other entries zero. Let $n \geq 2$. Please prove that $A \in M_n(\mathbb{R})$ is a linear combination of permutation matrices if and only if all its row sums and column sums equal the same number.

WORKED SOLUTION

Every permutation matrix has row and column sums equal to 1. Thus every linear combination has all those sums equal to the sum of its coefficients.
Conversely, let c be the common sum of A and put $B = A - cI_n$. All row and column sums of B are zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn},$$

where E_{ij} has one 1 in position (i, j) and zeros elsewhere. Then

$$B = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} b_{ij} F_{ij}.$$

The upper left blocks agree, and zero row and column sums determine the remaining entries. Each F_{ij} is the difference of two permutation matrices. Choose a permutation matrix P with ones in positions (i, j) and (n, n), matching the remaining $n - 2$ rows and columns in any order. Swap rows i, n to obtain a permutation matrix Q. Then

$$P - Q = E_{ij} + E_{nn} - E_{in} - E_{nj} = F_{ij}.$$

Thus every F_{ij} is a linear combination of permutation matrices. The identity is itself a permutation matrix, so $A = B + cI_n$ is a linear combination of them as well.

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